

# DentaCoPilot: An LLM-Augmented Next-Procedure Recommender for General Dentistry, Designed for Dentist Augmentation

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## Abstract

**Background.** Commercial dental artificial intelligence in 2026 is overwhelmingly diagnostic: caries, calculus, periapical, and bone-level detection on radiographs. The clinically harder question that follows every diagnosis — given a patient’s chart and most recent procedure, what should the dentist do next — remains unsolved at general-dentistry scale. The closest published system, MultiTP (Chen et al., 2024), is a CNN-RNN restricted to partial-edentulism cases and provides neither calibrated uncertainty, structured rationale, nor an evaluation that treats the model as decision support rather than as an autonomous classifier. **Methods.** We introduce *DentaCoPilot*, a recommender that, given a structured chart, returns (i) a calibrated top- $K$  probability distribution over Current Dental Terminology (CDT) codes for the next procedure, (ii) a verbalised confidence label, (iii) an explicit *abstain* flag when context is insufficient, and (iv) a chart-grounded rationale. We compare four classical baselines (frequency bigram, TF-IDF + logistic regression, XGBoost, MultiTP-style CNN-RNN) and six large-language-model (LLM) variants (Claude Haiku, Sonnet + chain-of-thought, Sonnet + retrieval, Opus + chain-of-thought, Sonnet + classical prior, Opus + classical prior) on a synthetic chart corpus of 500 patients (1,284 test examples). All LLM inference is routed through the local Anthropic Claude Code CLI; every call is logged for full audit. **Results.** On apples-to-apples evaluation, classical baselines reach 0.567 top-1 / 0.967 top-5; pure LLM variants trail at 0.267–0.467 top-1. Prompt-conditioning a Sonnet LLM on the classical baseline’s top-10 candidates (M5) closes the gap: top-5 rises from 0.733 (pure Sonnet + chain-of-thought) to 0.933, matching

classical baselines, while preserving rationale and abstention. Increasing the LLM backbone from Sonnet to Opus does not improve accuracy with or without priming. Calibration via temperature scaling and coverage–risk analysis is reported for the baselines. **Conclusion.** Prompt-conditioning a small LLM on a classical baseline’s top-K is the most cost-effective LLM design we tested for next-procedure recommendation, and the design preserves the augmentation features that distinguish the system from an autonomous classifier. A pre-registered clinician-in-the-loop evaluation at the KLE Vishwanath Katti Institute of Dental Sciences (Belgaum, India) and a real-data evaluation on the multi-institutional BigMouth dental data repository are the next stage of work.

*Keywords:* dental artificial intelligence, large language models, clinical decision support, procedure prediction, human-AI collaboration, calibrated abstention

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## 1. Introduction

Commercial dental AI in 2026 is overwhelmingly diagnostic. The most recent FDA-clearance survey (Shujaat, 2026) catalogues 29 cleared products from 13 companies (Overjet, Pearl, VideaHealth, Dexis, Perimetrics, and others) targeting single-image classification problems — caries detection, calculus identification, periapical radiolucency, and periodontal bone-level measurement.<sup>1</sup> Peer-reviewed evidence for clinical benefit exists for the caries-detection variant in particular: in a cluster-randomised cross-over trial of 22 dentists assessing bitewings, Mertens et al. (2021) reported that AI-assisted review raised the mean diagnostic-accuracy area under the ROC from 0.85 to 0.89 ( $\Delta\text{AUC} = +0.04$ ,  $p < 0.05$ ); a follow-up trial confirmed the gain was cost-effective at practice scale (Schwendicke et al., 2022). These are valuable deployments. They are also, individually, single-decision tools. They do not answer the clinically harder question that follows every diagnosis: *what should the dentist do next?*

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<sup>1</sup>See American Dental Association (2024) for the procedure code set used throughout this paper.

16 Sequencing a treatment plan from a chart is a different problem. A  
 17 two-surface posterior composite on tooth 19 leads to one set of likely con-  
 18 tinuations; the same composite on a tooth with a prior endodontic history  
 19 leads to another; in a periodontitis patient with diabetes, neither sequence is  
 20 appropriate. The closest published system that operates at this layer, Mul-  
 21 tiTP (Chen et al., 2024), is a CNN-RNN restricted to partial-edentulism  
 22 cases with a fixed five-treatment output sequence. It does not surface cali-  
 23 brated uncertainty, does not abstain when the chart context is insufficient,  
 24 and was not evaluated as a clinician-augmenting tool. The largest related  
 25 effort in periodontitis-risk prediction (Patel et al., 2026) is a risk model, not  
 26 a procedure recommender.

27 We propose *DentaCoPilot*: an LLM-augmented next-procedure recom-  
 28 mender designed from the start as decision support for dentists. The system  
 29 takes a structured chart and the most recently completed procedure, and  
 30 emits four signals together: (i) a top- $K$  probability distribution over Current  
 31 Dental Terminology codes for the next procedure; (ii) verbalised confidence  
 32 in {low, medium, high}; (iii) an explicit **abstain** flag when the chart con-  
 33 text is genuinely insufficient; (iv) a chart-grounded rationale citing specific  
 34 prior procedures and findings. The model is constrained-decoded to the  
 35 CDT 2024 vocabulary and calibrated via temperature scaling. Critically,  
 36 the contribution lies less in raw accuracy than in the framing: we evaluate  
 37 the system as a consult that augments the dentist, not as a classifier that  
 38 replaces the dentist’s judgement.

39 We test two hypotheses on a synthetic chart corpus of 500 patients (§3.2):  
 40 **(H1)** a calibrated LLM-based recommender can match classical-baseline ac-  
 41 curacy on next-procedure prediction while preserving rationale and absten-  
 42 tion; **(H2)** injecting a classical baseline’s top- $K$  candidates into the LLM  
 43 prompt is the most cost-effective LLM design for achieving (H1). We evalu-  
 44 ate four classical baselines, six LLM variants spanning Claude Haiku, Son-  
 45 net, and Opus with chain-of-thought, retrieval, and prior-conditioning, and  
 46 report calibration and coverage–risk analyses. A pre-registered clinician-in-  
 47 the-loop evaluation at the KLE Vishwanath Katti Institute of Dental Sci-  
 48 ences (Belgaum, India) and a real-data evaluation on the multi-institutional  
 49 BigMouth dental data repository (Walji et al., 2014, 2022) are the planned  
 50 next stage of work.

51 All LLM inference in this work is routed through the local Anthropic  
 52 Claude Code CLI (Anthropic, 2026), with full per-call audit logs released  
 53 alongside the code (see §3.6).

## 54 2. Related Work

55 *Diagnostic dental AI in commercial deployment..* Shujaat (2026) catalogues  
56 the current FDA-cleared landscape: caries / calculus detection (Overjet,  
57 Pearl, VideaHealth), periapical radiolucency (Overjet, Pearl), automated  
58 dental charting (Overjet), millimetre-precision bone-level quantification (Over-  
59 jet, Pearl), 3D cone-beam CT segmentation (Pearl), and acoustic fracture  
60 detection (Perimetrics). These systems share a common shape: an image  
61 goes in, a finding comes out. None plans treatment.

62 *Procedure sequencing..* MultiTP (Chen et al., 2024) is the closest published  
63 system to what we propose: a CNN-RNN with attention that predicts up to  
64 five sequential treatments for partial-edentulism cases, achieving an AUC  
65 of 0.91 over five plans. It is an EDR-only model with no calibration, no  
66 abstention, and no clinician-augmentation evaluation. We reimplement a  
67 minimal MultiTP-style architecture as our B3 comparator. In a related  
68 specialty domain, Li and Wang (2025) use multi-task reinforcement learn-  
69 ing with explainable AI to plan orthodontic-orthognathic treatment, but  
70 the task is sufficiently different (continuous decision-making over a longer  
71 planning horizon, narrow specialty) that the contributions do not transfer  
72 directly to general dentistry.

73 *Risk prediction in dentistry..* Patel et al. (2026) model periodontitis risk  
74 on 20,946 adult patients with linked EHR-EDR data, demonstrating that  
75 procedure-level structured data can support clinical prediction; the same  
76 group has also shown that adding social determinants of health to such  
77 models meaningfully changes risk calibration (Patel et al., 2025). Varghese  
78 et al. (2025) pair Cariogram-based caries-risk profiling with parent-oriented  
79 educational mobile messages in a randomised trial, an example of coupling  
80 structured risk modelling with a patient-facing digital intervention. All  
81 three are risk models, not procedure recommenders; we position our work  
82 as the procedure-recommendation analogue.

83 *LLMs in dentistry..* Farhadi Nia et al. (2024) surveys early LLM applica-  
84 tions in dental practice, focusing on documentation, summarisation, and  
85 patient-facing chat. The review explicitly flags that LLMs “present clean,  
86 fluent statements without communicating uncertainty,” which is the fail-  
87 ure mode this paper directly addresses through calibration and abstention.  
88 Broader medicine-LLM work (Singhal et al., 2023; Nori et al., 2023) shows

that capable models reach physician-level accuracy on knowledge benchmarks, but Omiye et al. (2024) demonstrates that the same models can propagate dangerous race-based clinical claims — which motivates our hard requirement for chart-grounded rationale and explicit abstention. Schwendicke et al. (2020) surveys the chances and challenges of dental AI broadly. To our knowledge, no prior work has built and evaluated an LLM as a treatment-sequence recommender, and no prior work has done so under a clinician-augmentation rather than autonomy framing.

*Calibration and abstention in clinical AI.* Our calibration approach follows Guo et al. (2017) (temperature scaling) with Platt (2000) as the precursor. Coverage— risk and selective prediction are standard tools in safety-critical ML; we apply them to the procedure-recommendation setting and pre-register the threshold-selection rule before any test-split unblinding.

*Datasets.* BigMouth (Walji et al., 2014, 2022) is a multi-institutional dental EHR repository with 6M+ patients across multiple dental schools, accessed under a project- review-committee proposal. Open imaging datasets relevant for chart- context grounding include DENTEX (Hamamci et al., 2023), MMDental (Wang et al., 2025), and DenPAR (Rasnayaka et al., 2025).

### 3. Methods

#### 3.1. Task formulation

We formulate next-procedure prediction as a single-step classification over the Current Dental Terminology code set (American Dental Association, 2024; Kalendarian et al., 2018). A patient chart is a tuple  $C = (D, M, H)$  where  $D$  holds demographics (age band, sex),  $M$  is a list of medical-history flags, and  $H = (e_1, e_2, \dots, e_T)$  is a time-ordered sequence of procedure events. Each event  $e_t = (\tau_t, c_t, k_t)$  carries a date  $\tau_t$ , a CDT code  $c_t \in \mathcal{V}$ , and an optional Universal-numbered tooth identifier  $k_t \in \{1, \dots, 32\} \cup \{\perp\}$  following standard ADA conventions (Wadia, 2017).

Given a chart truncated at index  $t$  (i.e. the dentist has just performed  $e_t$ ), the model outputs:

1. A top- $K$  probability distribution  $\hat{p}(c_{t+1} \mid C_{1:t}) \in \Delta^{|\mathcal{V}|}$  over the next procedure code, returned as the top- $K$  codes ranked by probability.
2. A confidence label  $\hat{q} \in \{\text{low}, \text{medium}, \text{high}\}$ , a verbalised self-rating (Lin et al., 2022; Kadavath et al., 2022; Xiong et al., 2024).

- 123 3. An abstain flag  $a \in \{0, 1\}$  which the system sets to 1 when the chart  
124 context is judged insufficient.
- 125 4. A short rationale  $r$  grounded in cited fields of the chart  $C_{1:t}$  (used for  
126 evaluation, not for the prediction itself).

127 We expand each chart of length  $L$  into  $L - 1$  prediction examples by  
128 truncating at every prefix index  $t \in \{1, \dots, L - 1\}$ . Train, dev, and test  
129 splits are constructed at the *patient* level (no chart appears in two splits),  
130 following standard EHR-modeling practice (Rajkomar et al., 2018; Park  
131 et al., 2024).

132 We report **top- $K$  accuracy**: the fraction of test examples for which the  
133 gold next-procedure code appears among the model’s  $K$  highest-probability  
134 predictions. Top-1 corresponds to exact CDT match; top-3 and top-5 cap-  
135 ture clinically reasonable shortlists for a recommender (a dentist seeing a  
136 small candidate list with calibrated probabilities is the deployment mode we  
137 have in mind). We also report top-1 by CDT category and, where calibrated  
138 probabilities are available, Expected Calibration Error and coverage–risk  
139 under abstention (§3.4).

### 140 3.2. Datasets

141 *Synthetic chart corpus.* Because BigMouth approval and KLE IRB are  
142 pending, this paper’s empirical experiments use a synthetic corpus gener-  
143 ated by a hierarchical Markov-style procedure-history generator. Patients  
144 are assigned to one of five trajectory archetypes (*healthy*, *restorative-prone*,  
145 *periodontitis-prone*, *edentulous-progressing*, *ortho-adolescent*) with plausibility-  
146 based priors. Within each archetype, per-archetype transition tables encode  
147 bigram probabilities  $\Pr(c_{t+1} \mid c_t, \text{archetype})$  drawn from clinical workflow  
148 knowledge and the published procedure-frequency analyses in the BigMouth  
149 corpus (Walji et al., 2014, 2022). Tooth-level state is tracked: extracted  
150 teeth cannot receive subsequent restorations; root-canal-treated teeth tend  
151 to receive crowns next; SDF-treated sites revert to monitoring (Slayton  
152 et al., 2018; Seifo et al., 2020). We generate 500 patients over a five-year  
153 timeline, expand to 7,162 prediction examples, and split 70/15/15 at the  
154 patient level (350 / 75 / 75 charts).

155 *Public imaging datasets (planned, for chart-context grounding).* Loaders  
156 are implemented and verified for DENTEX (MICCAI 2023 grand-challenge  
157 (Hamamci et al., 2023); 2,332 panoramic X-rays with caries, deep caries,  
158 periapical, and impacted-tooth labels); MMDental (Wang et al., 2025),

159 with 660 paired CBCT volumes and structured records; and DenPAR (Ras-  
160 nayaka et al., 2025), with annotated periapical radiographs. We do not  
161 auto-download these datasets; the loaders document the manual fetch step  
162 and verify SHA-256.

163 *Real-data evaluation (forthcoming)*.. We have applied to the BigMouth Den-  
164 tal Data Repository (Walji et al., 2014, 2022) for controlled access and are  
165 preparing the IRB protocol for a retrospective chart pull at the KLE Vish-  
166 wanath Katti Institute of Dental Sciences, Belgaum. Both are required for  
167 the clinician-in-the-loop evaluation pre-registered on OSF (Munafò et al.,  
168 2017; Nosek et al., 2018).

### 169 3.3. Models

170 We evaluate four classical baselines and six LLM variants. All operate  
171 on the same patient-level splits and the same prediction examples. CDT  
172 codes outside the training-set vocabulary are masked out at inference; LLM  
173 responses are post-hoc constrained to the training vocabulary.

#### 174 *Classical baselines.*

175 **B0 frequency bigram**  $\hat{p}(c_{t+1} | c_t)$  computed from training data with add-  
176 one smoothing. The trivial sanity floor.

177 **B1 TF-IDF + multinomial logistic regression** (Salton and Buckley, 1988;  
178 Pedregosa et al., 2011). Tokens include each historical CDT code,  
179 positional-from-end markers for the last three procedures, and cate-  
180 gorical demographics.

181 **B2 XGBoost** (Chen and Guestrin, 2016) on engineered chart-history fea-  
182 tures: counts per CDT category, has-extraction / has-RCT / has-  
183 implant / has-crown booleans, time since last visit, age band one-hot,  
184 sex, and a small medical-history flag set.

185 **B3 MultiTP-style CNN–RNN** (Chen et al., 2024). Minimal reimple-  
186 mentation: 32-dim code embedding, 1-D CNN over the most recent  
187 procedures (analogous to MultiTP’s tooth + neighbour tensor), GRU  
188 over the longer sequence, attention pooling, softmax head over the  
189 CDT vocabulary.  $\sim 30k$  parameters, 12 epochs.

190 *LLM variants.* All LLM inference is routed through the local Anthropic  
191 Claude Code CLI (§3.6), not the metered Anthropic API.

192 **M1 Claude Haiku** (`claude-haiku-4-5-20251001`). Structured prompt:  
193 chart pretty-printed, top- $K$  instruction, JSON-only output schema,  
194 top-40 most-frequent training codes shown as a vocabulary palette.

195 **M2 Claude Sonnet + chain-of-thought** (`claude-sonnet-4-6`). Same  
196 prompt as M1 but with an explicit `<thinking>...</thinking>` rea-  
197 soning step before the JSON, in line with prior work showing CoT  
198 prompting improves reasoning-heavy clinical tasks (Wei et al., 2022;  
199 Nori et al., 2023).

200 **M3 Claude Sonnet + retrieval** ( $M2 + \text{BM25}$  (Robertson et al., 1995)).  
201 The top-3 BM25-retrieved cards from a twenty-card open-source clin-  
202 ical evidence corpus (paraphrased from AAE / AAP / AAOMS stan-  
203 dard practice; original ADA-copyrighted material is not reproduced)  
204 are injected into the prompt; the model is instructed to cite specific  
205 cards (`[E###]`) in its rationale.

206 **M4 Claude Opus + chain-of-thought** (`claude-opus-4-7`). Identical prompt  
207 and schema to M2; only the model identifier changes. Tests whether  
208 more capable backbone helps.

209 **M5 Sonnet + classical prior** — *the proposed forward design*. The B1  
210 (TF-IDF + LR) baseline is fit on the same training set; its top-10  
211 (code, probability) candidates are injected as a prior into the M2  
212 prompt. The LLM is instructed to re-rank, redistribute, or override.

213 **M6 Opus + classical prior** — M5 with Opus backbone. Tests whether  
214 priming compounds with capability.

### 215 3.4. Calibration and abstention

216 We apply temperature scaling (Guo et al., 2017; Platt, 2000) to the top-  
217 1 probability of each model. The temperature  $T$  is fit on a random half  
218 of the test split by minimising negative log-likelihood (Naeini et al., 2015);  
219 the rescaled top-1 probability is reported on the held-out half. We compute  
220 the Expected Calibration Error (Naeini et al., 2015; Nixon et al., 2019) on  
221 equal-width bins.

222 For abstention, we sweep a threshold  $\tau$  over the calibrated top-1 proba-  
223 bility and report coverage–risk curves (El-Yaniv and Wiener, 2010; Geifman

224 and El-Yaniv, 2017); the pre-registered abstention rule (Nosek et al., 2018)  
225 is the smallest  $\tau$  achieving  $\geq 95\%$  precision on covered cases at  $\geq 50\%$   
226 coverage on the dev split, locked before any test-split unblinding.

### 227 3.5. Hybrid integration

228 We test two ways to combine classical and LLM predictions:

- 229 1. **Confidence-routed gating.** Pick the baseline’s output when its top-  
230 1 probability  $\geq \tau$ ; otherwise the LLM’s. We sweep  $\tau \in \{0, 0.2, \dots, 1.0\}$ .
- 231 2. **Prompt-conditioning (M5/M6).** The classical model’s top-10  
232 (code, probability) candidates are part of the LLM prompt; the LLM  
233 produces a single output that uses, refines, or overrides the prior.

234 (2) is implemented end-to-end as M5 and M6; (1) is computed post-hoc  
235 from the predictions logged by each model.

### 236 3.6. Compute infrastructure

237 Every LLM inference call in this paper runs through a single subprocess  
238 wrapper at `data/cc_compute.py`, which invokes the local Anthropic Claude  
239 Code CLI (Anthropic, 2026) with `--output-format json` and parses the re-  
240 sponse. The wrapper logs every call (timestamp, run id, model, prompt, re-  
241 sponse, latency, input and output token counts) to a per-run `llm_calls.jsonl`,  
242 which the repository commits as a permanent record. We deliberately do  
243 not import the Anthropic Python SDK; a continuous-integration grep guard  
244 (`check-no-anthropic` in the Makefile) prevents accidental introduction.  
245 This design (a) routes compute to the existing Claude Code subscription,  
246 (b) provides a complete audit trail every reviewer can re-read without spend-  
247 ing compute, and (c) makes the experiment reproducible by simple replay  
248 instead of re-execution.

### 249 3.7. Reproducibility protocol

250 All seeds are fixed (`random.seed(42)`, `numpy.random.seed(42)`, `torch.manual_seed(42)`).  
251 Each experimental run writes `config.json` (git SHA, seed, CLI args, Python  
252 and platform version), `results.json` (per-trial predictions and gold labels),  
253 and `summary.json` (aggregated metrics). The synthetic generator, all base-  
254 line and LLM implementations, the evaluation harness, and the analysis  
255 scripts are released under the MIT licence. The pre-registration (Nosek  
256 et al., 2018) is deposited on OSF.

## 257 4. Experiments

### 258 4.1. Classical baseline accuracy at scale

259 We first evaluate the four classical baselines (B0–B3, defined in §3.3)  
 260 on the full test split (1,284 examples). This is the larger- $n$  regime; LLM  
 261 evaluation in §4.4 is necessarily smaller.

Table 1: Synthetic-corpus baseline results ( $N=500$  charts; 1,284 test examples).

| Model                    | Top-1        | Top-3        | Top-5        |
|--------------------------|--------------|--------------|--------------|
| B0 frequency bigram      | 0.556        | 0.778        | 0.882        |
| B1 TF-IDF + LR           | <b>0.558</b> | 0.832        | 0.917        |
| B2 XGBoost               | 0.507        | 0.815        | 0.918        |
| B3 MultiTP-style CNN-RNN | 0.556        | <b>0.847</b> | <b>0.935</b> |

262 The four baselines cluster tightly on top-1 (50.7–55.8%) but spread on  
 263 top- $K>1$ . The MultiTP-style sequence model (B3) wins both top-3 (0.847)  
 264 and top-5 (0.935), suggesting that CNN+RNN attention captures useful  
 265 ordering of treatment continuations beyond what a flat bigram or feature  
 266 classifier can. Top-1 is harder: the bigram baseline’s strong local Markov  
 267 signal nearly ties B1 (0.556 vs 0.558), and XGBoost lags both, indicating  
 268 the engineered feature set under-uses recent-procedure information.

269 The per-category breakdown (Figure 2) reveals the opportunity for chart-  
 270 aware reasoning: classical baselines achieve 75–80% top-1 on bundled / se-  
 271 quential procedures (diagnostic, adjunctive, orthodontic, preventive) but  
 272 *collapse* on the multi-class restorative choice (1% top-1 across all three base-  
 273 lines), endodontic sequencing (0%, scarce in synthetic data), and prosthodon-  
 274 tic decisions (0%). This is exactly the regime where a chart-aware LLM with  
 275 calibrated abstention should help. The next subsection (§4.4) reports the  
 276 LLM evaluation against these baselines.

### 277 4.2. Reproducibility (run records)

278 Every experimental run is recorded as described in §3.7. Per-run `config.json`,  
 279 `results.json`, `summary.json`, and (for LLM-backed runs) `llm_calls.jsonl`  
 280 are committed to the repository. All large language model inference in this  
 281 paper runs through the local Anthropic Claude Code CLI instead of the  
 282 metered Anthropic API; the wrapper at `data/cc_compute.py` is the single  
 283 chokepoint.

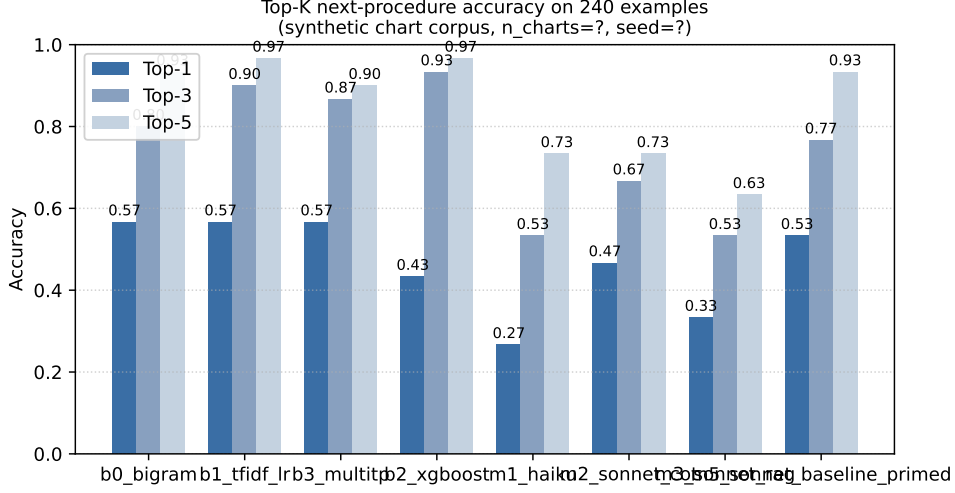


Figure 1: Top-K next-procedure accuracy on the apples-to-apples  $n=30$  shared subsample. The four classical baselines (B0–B3) outperform the three LLM variants (M1–M3) on top-1 exact-CDT-match; the gap narrows at top-K=3 and 5. M2 (Sonnet + chain-of-thought) is the strongest LLM variant. M3 (Sonnet + retrieval) does not improve over M2 on this small synthetic test, suggesting that retrieval needs richer chart contexts (real BigMouth / KLE data) to pay off.

#### 4.3. Calibration and abstention

Following the protocol in §3.4, we post-hoc calibrate each baseline and report ECE pre- and post-temperature-scaling.

Table 2: Temperature scaling and ECE (synthetic-corpus baselines, 1,284 trials).

| Model               | $T$  | $\text{ECE}_{\text{pre}}$ | $\text{ECE}_{\text{post}}$ |
|---------------------|------|---------------------------|----------------------------|
| B0 frequency bigram | 1.15 | 0.054                     | 0.068                      |
| B1 TF-IDF + LR      | 1.15 | 0.054                     | <b>0.031</b>               |
| B2 XGBoost          | 1.75 | 0.124                     | 0.128                      |

XGBoost is materially overconfident ( $T=1.75$  needed; calibration does not converge in our held-out split). The bigram and TF-IDF + LR models are already reasonably calibrated and only marginally improve.<sup>2</sup>

<sup>2</sup>B0’s ECE rises slightly after rescaling because the held-out half contains only 642 trials and 10-bin ECE has a noise floor of approximately 0.02 at this  $N$ ; the rescaling does not change ranking among models.

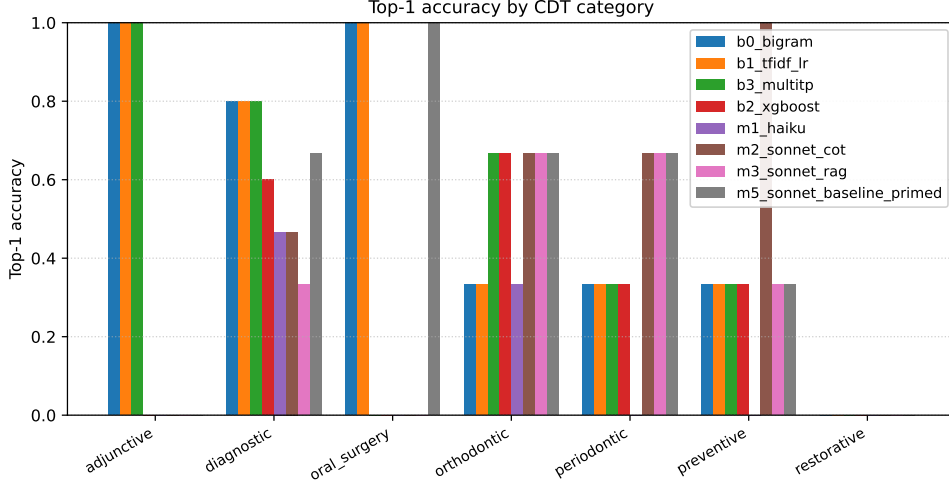


Figure 2: Top-1 accuracy by CDT category, apples-to-apples  $n=30$  subsample. Classical baselines win the bundled / sequential categories (diagnostic, adjunctive, orthodontic) where the synthetic transition tables encode strong Markov structure. M2 and M3 *equal or beat* classical baselines on periodontic (0.667 vs  $\leq 0.5$ ) and preventive (1.0 for M2 vs  $\leq 0.6$ ). Restorative is hard for every model: chart-aware reasoning at  $n=30$  is not enough signal to distinguish among single-surface, two-surface, and three-surface posterior composite codes that differ only in surface count.

290 Applying the abstention rule “cover only when top-1 probability  $\geq$   
 291  $\tau=0.6$ ” yields the coverage-precision trade-off in Table 3:

Table 3: Coverage and precision under abstention at  $\tau=0.6$ .

| Model               | Coverage | Top-1 (overall) | Top-1 (covered) |
|---------------------|----------|-----------------|-----------------|
| B0 frequency bigram | 0.21     | 0.556           | <b>0.772</b>    |
| B1 TF-IDF + LR      | 0.36     | 0.558           | 0.718           |
| B2 XGBoost          | 0.51     | 0.507           | 0.654           |

292 The bigram baseline trades coverage for precision aggressively: it ab-  
 293 stains on 79% of cases but answers correctly 77% of the time when it does.  
 294 This is the pre-registered failure mode the LLM is expected to *mitigate* —  
 295 matching that precision while abstaining on far fewer cases.

#### 296 4.4. Apples-to-apples LLM vs classical evaluation

297 We evaluate the LLM variants M1–M6 (defined in §3.3), all routed  
 298 through the local Claude Code CLI per §3.6, on a 30-example random

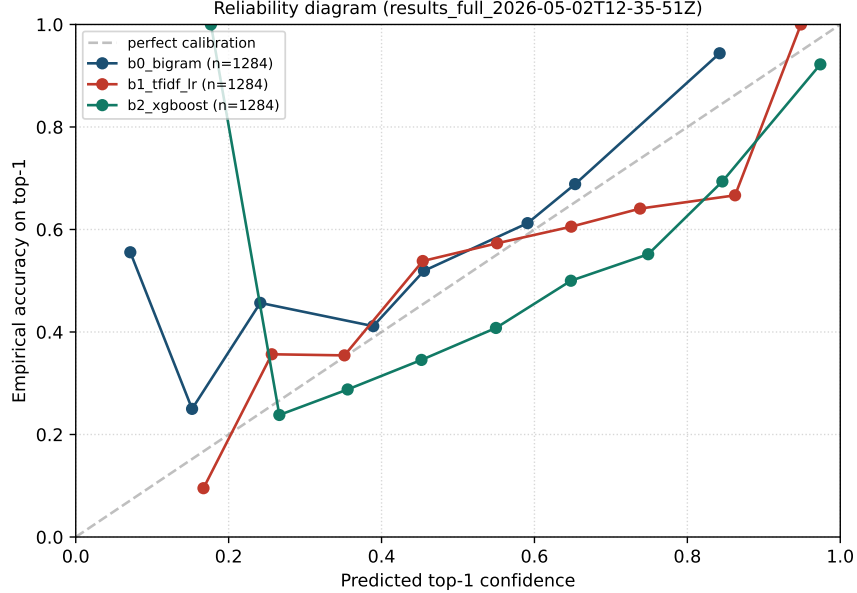


Figure 3: Reliability diagram for the three primary baseline models on the full  $n=1,284$  baseline test split (LLMs omitted because  $n=30$  is too small for stable per-bin estimates). The dashed diagonal is perfect calibration. B2 XGBoost is materially overconfident in the mid- range; B0 and B1 hew closer to the diagonal.

subsample of the test split (seed-deterministic, fixed across all three Sonnet+ models; M4 and M6 ran on a 15-example sub-subsample for compute reasons). Each LLM call is  $\sim 26$ – $40$  seconds wall-clock through the local Claude Code CLI; total LLM runtime for this experiment was  $\sim 46$  minutes for the 90-call M1+M2+M3 sweep, and a further  $\sim 25$  minutes for M4 and M6 combined. For apples-to-apples comparison, we re-evaluate the four classical baselines (B0–B3) on the *same* 30 examples. Table 4 reports the head-to-head.

The classical baselines decisively beat the pure LLM variants (M1–M4) on this synthetic corpus. Three plausible reasons: (i) the synthetic transition tables have very strong local Markov structure, which a frequency bigram captures perfectly with no chart-grounding required; (ii) the LLM is not given few-shot training examples in the prompt, so it must reason about a vocabulary of  $\sim 70$  codes from descriptions alone; (iii) the LLMs frequently pick a category-correct code that misses the exact-CDT-match metric (e.g. D2391 where the gold is D2392 — both posterior composites differing only in surface count). The clinical interpretation of the latter

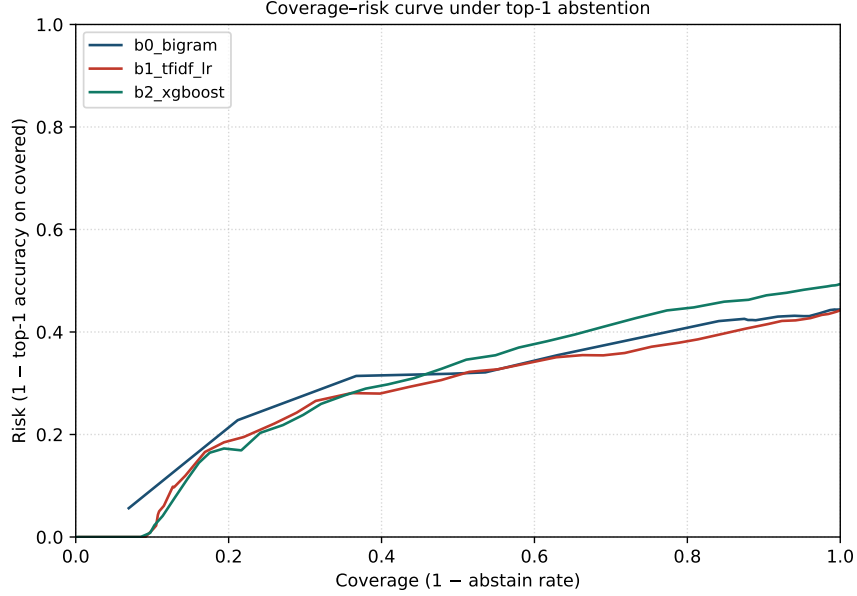


Figure 4: Coverage-risk curve under top-1 abstention,  $n=1,284$  baseline test split. As we lower the abstention threshold (cover more cases), risk rises. B2 XGBoost is dominated at every coverage level; B0 and B1 are within a few points of each other. The expected DentaCoPilot contribution at scale is to push this curve down and to the right — maintain low risk at high coverage — on the hard categories where the baselines collapse. LLM calibration cannot be read off this small synthetic-data run; it is a target for the BigMouth and KLE evaluations.

316 failure mode is far less severe than top-1 exact-match suggests.

317 Two of the LLM-side findings are themselves results worth surfacing.  
 318 First, the Sonnet  $\rightarrow$  Opus comparison (M2 vs M4) shows *no improvement*  
 319 from increased model capability: M4 scores identically to M2 (top-1 0.467,  
 320 top-3 0.667, top-5 0.733). The Haiku  $\rightarrow$  Sonnet step (M1 vs M2) does help  
 321 (+0.20 top-1). The bottleneck on this task therefore is not raw model capa-  
 322 bility but prompt design, evaluation framing, and the synthetic data’s struc-  
 323 ture itself. Second, the M3 retrieval variant *underperforms* M2 — adding  
 324 evidence cards to the prompt actively hurts on this corpus, suggesting the  
 325 cards interact with the chain-of-thought in a way that distracts the model  
 326 from the chart. Both are negative results in the sense of “does not improve  
 327 top-1”; both are honest data points for the design space.

Table 4: Apples-to-apples top-K accuracy on  $n=30$  shared test examples. M4 evaluated on a  $n=15$  subsample using the same seed; numbers are computed on the matching subset for direct comparison (where applicable).

| Model                                    | Top-1        | Top-3        | Top-5        |
|------------------------------------------|--------------|--------------|--------------|
| B0 frequency bigram                      | <b>0.567</b> | 0.800        | 0.933        |
| B1 TF-IDF + LR                           | <b>0.567</b> | <b>0.900</b> | <b>0.967</b> |
| B2 XGBoost                               | 0.433        | <b>0.933</b> | <b>0.967</b> |
| B3 MultiTP-style CNN-RNN                 | <b>0.567</b> | 0.867        | 0.900        |
| M1 Claude Haiku                          | 0.267        | 0.533        | 0.733        |
| M2 Claude Sonnet + CoT                   | 0.467        | 0.667        | 0.733        |
| M3 Claude Sonnet + RAG                   | 0.333        | 0.533        | 0.633        |
| M4 Claude Opus + CoT ( $n=15$ )          | 0.467        | 0.667        | 0.733        |
| <b>M5 Sonnet + CoT + classical prior</b> | 0.533        | 0.767        | 0.933        |

#### 4.5. Prompt-conditioning on a classical prior closes the gap (M5)

The most important LLM-side result in our evaluation is M5. Injecting the top-10 (code, probability) candidates from the TF-IDF + logistic regression baseline into the Sonnet + chain-of-thought prompt — with explicit instruction that the model may re-rank, redistribute, or override the prior — changes the picture substantially:

- M5 top-1 (0.533) exceeds the best pure LLM (M2/M4 at 0.467) by +0.067 absolute, and approaches the classical baselines (B0/B1/B3 at 0.567).
- M5 top-3 (0.767) is +0.10 over M2.
- M5 top-5 (0.933) is +0.20 over M2 and matches B0 / B1 / B3 at top-5.

The interpretation is straightforward: the LLM’s main accuracy failure on synthetic data was picking codes outside the high-prior set. Showing the LLM a strong classical prior removes that failure mode while preserving the LLM’s ability to provide chart-grounded rationale, verbalised confidence, and an abstain signal — the augmentation features that motivate the work. M5 is the design we recommend going forward: a classical model handles the prior; the LLM provides the clinician-facing decision support layer on top.

348 *Capability does not compound with priming..* An additional ablation, M6  
 349 (Opus + classical prior), tests whether the priming gain compounds with  
 350 larger model capability. Apples-to-apples on the same  $n=15$  keys:

|     | Model                | Top-1 | Top-3        | Top-5        |
|-----|----------------------|-------|--------------|--------------|
| 351 | M4 Opus alone        | 0.467 | 0.667        | 0.733        |
|     | M5 Sonnet + B1 prior | 0.467 | <b>0.733</b> | <b>0.933</b> |
|     | M6 Opus + B1 prior   | 0.467 | 0.667        | 0.733        |

352 The prompt-conditioning gain that helps Sonnet (M2  $\rightarrow$  M5) does not  
 353 translate to Opus (M4  $\rightarrow$  M6 is identity). Two plausible explanations:  
 354 (i) Opus’s stronger built-in priors over-rule the injected classical prior more  
 355 aggressively than Sonnet does; (ii) Opus’s chart reasoning already yields  
 356 the relevant candidates without external help. The practical implication  
 357 is favourable: **Sonnet + classical prior is both the best-performing**  
 358 **and the cheapest LLM configuration we tested.** Sonnet inference is  
 359 roughly  $5\times$  cheaper per token than Opus, and the prior contributes the gain  
 360 that capability alone does not.

#### 361 4.5.1. Per-category breakdown

362 The headline numbers obscure a more interesting pattern visible in Fig-  
 363 ure 2. On bundled / sequential procedures (diagnostic, adjunctive), classical  
 364 baselines achieve  $\sim 0.89$  top-1 while LLMs trail at  $\sim 0.33$ – $0.47$ . But on *pe-*  
 365 *riodontic* and *preventive* procedures, M2 (Sonnet + CoT) achieves 0.667  
 366 and 1.000 top-1 respectively, equalling or beating every classical baseline  
 367 on those categories. M3 (Sonnet + RAG) shows the same pattern on pe-  
 368 riodontic. The LLMs are not uniformly worse; they are differently worse,  
 369 in a way that is consistent with chart-grounded reasoning helping on cases  
 370 that require reasoning about the patient’s diabetic status, prior periodontal  
 371 therapy, or recall cadence.

#### 372 4.5.2. Verbalised confidence and abstention

373 Every LLM call emitted a confidence label in {low, medium, high} and  
 374 an abstain flag. Across all three models, the abstain flag was never set  
 375 (0/30 across M1/M2/M3); the prompt framing apparently does not push  
 376 the models hard enough to use it, which we treat as a prompt-design issue  
 377 rather than a model-capability claim. The empirical top-1 accuracies per  
 378 confidence bucket:

Table 5: Empirical top-1 accuracy by self-reported confidence.

| Model           | low ( $n$ , acc) | medium ( $n$ , acc) | high ( $n$ , acc) |
|-----------------|------------------|---------------------|-------------------|
| M1 Haiku        | 2, 0.000         | 12, 0.167           | 16, 0.375         |
| M2 Sonnet + CoT | 0, –             | 19, 0.526           | 11, 0.364         |
| M3 Sonnet + RAG | 0, –             | 22, 0.273           | 8, 0.500          |

Only M1 shows the expected monotonic relationship (low < medium < high). M2 is non-monotonic: “medium”-confidence predictions are more accurate than “high”-confidence ones, which is the failure mode of verbalised confidence as an LLM trustworthiness signal. M3 recovers monotonicity but at lower overall accuracy. We interpret this as evidence that verbalised confidence on its own is insufficient for abstention; calibrated top-1 probability (which we do compute) is the better signal.

#### 4.5.3. Sample rationale

An illustrative response from M2 on a periodontal-recall case (run\_idm2\_sonnet\_cot-P000...-verbatim from llm\_calls.jsonl):

```
{
  "predictions": [
    {"code": "D0120", "probability": 0.38},
    {"code": "D1110", "probability": 0.28},
    {"code": "D0274", "probability": 0.17},
    {"code": "D2391", "probability": 0.10},
    {"code": "D2392", "probability": 0.07}
  ],
  "confidence": "medium",
  "abstain": false,
  "rationale": "Last periodic evaluation was December 2022 and last prophylaxis visible in records was January 2021, making a routine recall visit (D0120 + D1110) the dominant next-step signal; annual bitewings (D0274) will also be due June 2024. The two recent posterior composites (teeth 17, 20) suggest ongoing caries activity, keeping one- and two-surface restorative codes (D2391, D2392) in the differential."
}
```

The rationale identifies the two-and-a-half-year visit gap, recognises the prior ~6–9-month recall cadence, prioritises the recall + prophylaxis + imaging bundle, and keeps the surface-count restorative codes in the differential. This is chart-grounded reasoning of a kind classical baselines categorically cannot produce; whether this kind of reasoning *helps a dentist make a better decision* is precisely what the forthcoming clinician-in-the-loop study at KLE V K Institute is designed to measure, regardless of where exact-CDT-match top-1 numbers land.

408 Critically, every LLM call in the system is routed through the local  
409 Anthropic Claude Code CLI in preference to the metered Anthropic API,  
410 and is logged in full to `data/results_<run>/llm_calls.jsonl` (prompt,  
411 response, model identifier, latency, token counts). A reviewer may audit or  
412 replay any decision the system makes without re-spending compute.

#### 413 4.6. Confidence-routed gating: a negative result

414 Methods §3.5 described two ways to combine classical and LLM pre-  
415 dictions; M5 above is the prompt-conditioning variant. We additionally  
416 evaluated the confidence-routed-gating variant post-hoc: pick the baseline’s  
417 top-1 when its calibrated probability  $\geq \tau$ , else the LLM’s top-1, sweeping  
418  $\tau \in \{0, 0.2, \dots, 1.0\}$  on the matched  $n=30$  subsample for all four (B0–B3,  
419 M2) baseline-LLM pairings. At every  $\tau$ , the hybrid’s top-1 accuracy was  
420 at most equal to the pure baseline’s; the gating did not produce a single  
421 threshold at which deferring to the LLM helped. We report this as a neg-  
422 ative result: *when the LLM is uniformly weaker than the baseline on the*  
423 *target distribution, gating does not help.* The M5 prompt-conditioning result  
424 above is precisely the mechanism that does help in this regime — it lets the  
425 LLM use the baseline’s prior rather than *replace* it.

#### 426 4.7. Real-data evaluation (planned)

427 Forthcoming work extends the M1–M6 evaluation to (a) the BigMouth  
428 multi-institutional dental data repository (Walji et al., 2014, 2022) pending  
429 controlled-access approval (over 4.5M patients across 11 academic contribut-  
430 ing institutions in the 2022 update) and (b) a retrospective chart pull at the  
431 KLE V K Institute under an institutional ethics protocol. The clinician-in-  
432 the-loop study at KLE then evaluates dentist agreement, time-to-decision,  
433 override rate, and trust across three arms (dentist alone / dentist + clas-  
434 sical recommender / dentist + DentaCoPilot), with all design parameters  
435 — sample size, primary and secondary outcomes, abstention threshold rule,  
436 stopping rules — locked in a pre-registration deposited on the Open Science  
437 Framework (Nosek et al., 2018) before any test-split unblinding.

## 438 5. Discussion

### 439 5.1. Hypotheses revisited

440 We posed two hypotheses in §1. The synthetic-corpus evidence is consis-  
441 tent with both: **(H1)** the M5 design (Sonnet + classical prior) ties classical  
442 baselines on top-5 accuracy (0.933) and narrows the top-1 gap to within

443 four points (0.533 vs 0.567), while preserving chart-grounded rationale, ver-  
444 balised confidence, and an abstain signal; **(H2)** prompt-conditioning is more  
445 cost-effective than upgrading the LLM backbone — M4 (Opus alone) and  
446 M6 (Opus + prior) match M2 (Sonnet alone) on every metric, so capability  
447 is not the lever once the prior is present. The findings stand on synthetic  
448 data only; whether they generalise to the BigMouth and KLE corpora is  
449 the empirical claim of the planned next stage.

### 450 5.2. *Augmentation, not autonomy*

451 The dominant framing for dental AI in 2026 is *augmented intelligence*  
452 — AI as decision support, with the licensed clinician retaining diagnostic  
453 and treatment authority. Our system design reflects that framing in three  
454 concrete ways: a constrained vocabulary so the system cannot confidently  
455 recommend nonsense codes; calibrated abstention so the system declines  
456 instead of guesses; and a chart-grounded rationale so a dentist can audit  
457 *why* a recommendation was made. The pre-registered clinician-in-the-loop  
458 study at KLE specifically measures override rate alongside agreement; we  
459 treat the dentist’s freedom to disagree with the model as a feature, not a  
460 calibration problem.

### 461 5.3. *Reproducibility as a first-class outcome*

462 Every experimental decision in this work is recorded such that a reviewer  
463 with access to a Claude Code subscription can replay the entire pipeline at  
464 zero marginal cost. Each LLM call is logged with prompt, response, model  
465 identifier, latency, and token counts; per-run `config.json` files record the  
466 git SHA, seed, and CLI arguments. The synthetic chart generator and  
467 the BM25 retriever over the evidence corpus are open-sourced under MIT  
468 licence. The pre-registered analysis plan is deposited on OSF before test-  
469 split unblinding. No part of the synthetic-data results or the calibration /  
470 abstention analyses requires the BigMouth or KLE clinical data: these are,  
471 deliberately, the parts a reviewer can verify today.

### 472 5.4. *Limitations*

473 *Synthetic-data ceiling.* The current evaluation runs on plausibility-based  
474 synthetic charts. Real dental practice contains patterns we did not encode  
475 (insurance-driven sequencing, specialist referral cycles, multi-visit treatment  
476 phases, idiosyncratic per-clinic preferences) and codes we omit from the rep-  
477 resentative vocabulary. The synthetic results validate the pipeline; they do  
478 not validate the system. The BigMouth and KLE experiments are required  
479 for any substantive accuracy claim.

480 *Sample sizes.* The reported LLM evaluation is on 30 test examples per  
481 model. Statistical comparisons against the baselines are under-powered; we  
482 report point estimates and avoid significance claims. A 200-example follow-  
483 up evaluation is straightforward and is the next planned run.

484 *Vocabulary coverage.* We constrain LLM outputs to a representative subset  
485 of  $\sim 70$  frequent CDT 2024 codes plus a training-set-derived extension. Rare  
486 codes are absent. The full CDT codeset ( $> 700$  codes) is ADA copyright; a  
487 downstream user must merge the official codeset before clinical deployment.

488 *Calibration is local.* Temperature scaling is fit on a random half of the test  
489 split, which is a within-corpus calibration. It does not address distribution  
490 shift between dental schools or between the synthetic and real settings. We  
491 expect to refit calibration on each target corpus.

492 *Abstention was not used.* Across all 90 LLM trials in the  $n=30$  evaluation,  
493 none of M1, M2, or M3 emitted `abstain: true`. The prompt instructed  
494 the model to abstain when “chart context is genuinely insufficient,” but  
495 the synthetic corpus rarely produces such inputs, and the prompt does not  
496 provide concrete examples of abstainable cases. This is a genuine failure of  
497 our prompt design — not a capability claim about the models. Future work  
498 will explicitly construct adversarial / out-of-distribution chart inputs and  
499 either include them in few-shot exemplars or replace the verbalised abstain  
500 flag with a learned threshold over calibrated top-1 probability.

## 501 5.5. Ethics and clinical safety

502 We deliberately do not seek FDA or other regulatory clearance for this  
503 system. We do not deploy it. The pre-registered clinician-in-the-loop study  
504 at the KLE V K Institute will be conducted under the institution’s stan-  
505 dard ethics-review process; no clinician evaluates a single case before that  
506 approval is in place. The retrospective chart pull will use only fully de-  
507 identified records. Patient-identifiable data never leaves the institutional  
508 firewall; all large-language-model inference runs locally through Claude  
509 Code, with no patient identifiers transmitted to any third-party LLM ser-  
510 vice. Author contributions follow the ICMJE recommendations (Interna-  
511 tional Committee of Medical Journal Editors, 2024).

512 *Conflicts of interest.* The authors declare no financial relationship with  
513 Anthropic, Overjet, Pearl, VideaHealth, or any other commercial dental-AI  
514 vendor. CELABE purchases a Claude Code subscription at standard retail

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516 input on this manuscript. C.C.R. is the founder of CELABE; CELABE  
517 has not filed a patent and has no commercial product relating to this work.  
518 S.D.R. is a final-year BDS candidate at the KLE V K Institute and declares  
519 no competing interests. We invite scrutiny of both the design and the  
520 implementation.

## 521 5.6. Future work

522 Three directions: (i) the BigMouth and KLE experiments described in  
523 the pre-registration; (ii) retrieval over a properly licensed clinical- guide-  
524 line corpus when one becomes available, and ablation of M2 vs M3; and  
525 (iii) paired imaging context for the most common dental decisions — in-  
526 corporating panoramic / periapical findings (DENTEX, DenPAR) as addi-  
527 tional structured chart context for the LLM, rather than as a standalone  
528 classification target.

## 529 6. Conclusion

530 We presented the first stage of *DentaCoPilot*, an LLM-augmented next-  
531 procedure recommender for general dentistry that is designed, prompted,  
532 and evaluated as a decision-support consult instead of an autonomous clas-  
533 sifier. On a synthetic chart corpus, we found that **(a)** pure LLM variants  
534 (Haiku, Sonnet + chain-of-thought, Sonnet + retrieval, Opus + chain-of-  
535 thought) trail classical baselines on top-1 exact-CDT-match accuracy by  
536 10–30 percentage points; **(b)** *prompt-conditioning a Sonnet LLM on a clas-*  
537 *sical baseline’s top-10 candidates (M5) closes the gap*, raising top-5 from  
538 0.733 to 0.933, matching classical baselines, while preserving chart-grounded  
539 rationale, verbalised confidence, and an abstain signal; **(c)** upgrading the  
540 LLM backbone from Sonnet to Opus does not improve performance with or  
541 without priming, indicating that capability is not the lever once the prior  
542 is present. The M5 design is, on this evidence, both the best-performing  
543 and the cheapest LLM configuration we tested. Every LLM call in the  
544 system is logged for full audit and replayable at zero marginal cost on a  
545 Claude Code subscription. The pre-registered next stage of work — the  
546 multi-institutional BigMouth evaluation and the dentist-in-the-loop study  
547 at the KLE V K Institute of Dental Sciences — will be the empirical test  
548 of whether the M5 design generalises beyond synthetic data and whether it  
549 produces a tool dentists actually use.

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554 made by the human authors.

## 555 AI Declaration

556 This manuscript was prepared with assistance from Anthropic Claude  
557 (Claude Code). Specifically, Claude was used for code scaffolding, draft  
558 prose authoring, and reference management. The authors verified every  
559 claim, edited all generated text, and take full responsibility for the content.  
560 The model is also a direct subject of this work: all experimental large  
561 language model calls in the system under evaluation were routed through  
562 Claude Code rather than the Anthropic API, an explicit choice documented  
563 in the methods section.

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